

DHCP & DNS

1. 3 way- handshakes

The **3-way handshake** is a fundamental process in the **TCP (Transmission Control Protocol)** used to establish a reliable connection between a client and a server. This handshake ensures that both parties are ready to send and receive data before the communication starts.

1. SYN (Synchronize)

The client initiates the connection by sending a SYN (synchronize) packet to the server.

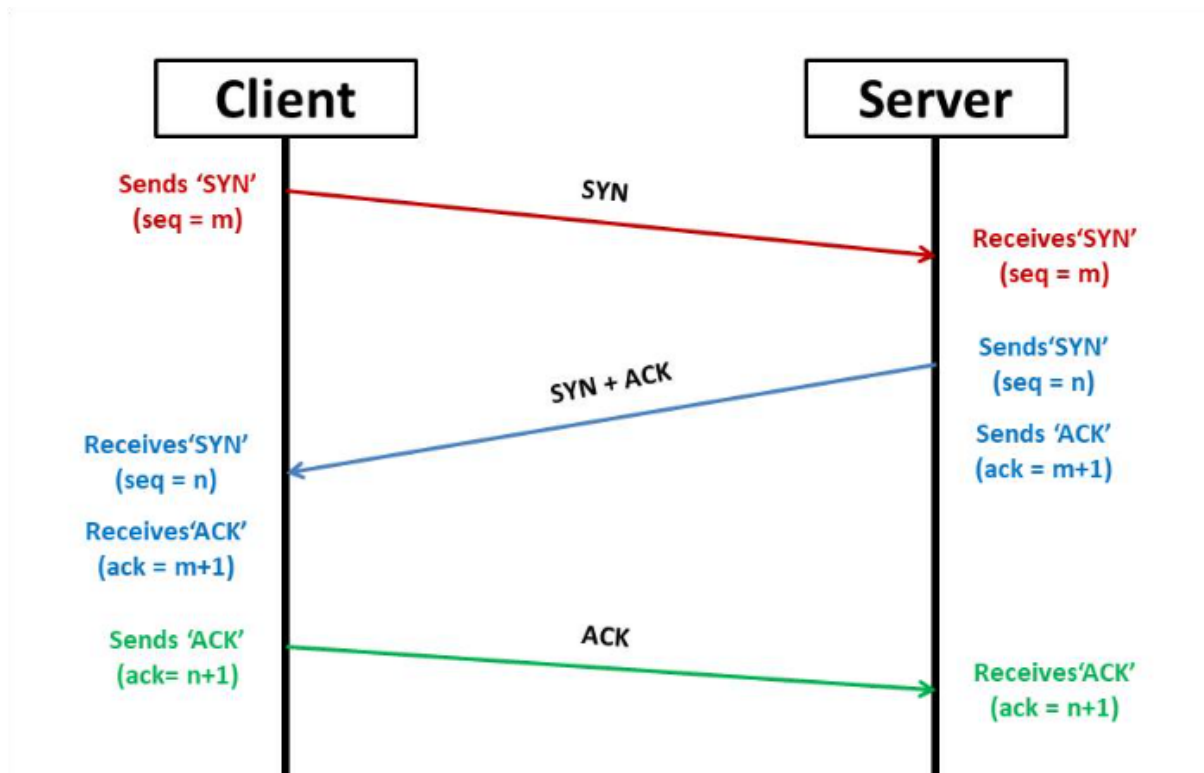
2. SYN-ACK (Synchronize-Acknowledge)

The server responds to the client with a **SYN-ACK** packet:

3. ACK (Acknowledge)

The client sends an **ACK** packet back to the server:

Client	Server
---- SYN Seq=X ----->	
<--- SYN-ACK Seq=Y Ack=X+1 -----	
---- ACK Seq=X+1 Ack=Y+1 ----->	



3-Way Handshaking(for establishing connection)

three steps to establish the connection in 3-way handshakes:

1. Client sends SYN.
2. Server responds with SYN-ACK.
3. Client confirms with ACK.

2. DHCP (Dynamic Host Configuration Protocol)—

Assigns automatically IP Addresses in the network.

DHCP is a **client-server protocol** that automates the assignment of:

1. **IP Addresses** (from a predefined range or pool)
2. **Subnet Masks**
3. **Default Gateways**
4. **DNS Server addresses**
5. **Other optional parameters**, like domain names.

Without DHCP, these parameters would have to be manually configured, which is time-consuming and error-prone, especially for large networks.

DHCP Server-

Windows Server/ Linux Machines / Cisco Router/Switches.

DHCP Client -

PC/Laptop/Mobile

(IPv4 address from a **pool of addresses** for a limited period. The lease is typically from 24 hours to a week or more.)

How DHCP Works

The DHCP process involves four main steps, often referred to as **DORA**:

1. D (Discover) :

- The client device (host) sends a **DHCP Discover broadcast** (255.255.255.255) to find available DHCP servers.
- This packet is sent because the client doesn't yet have an IP address.

2. O (Offer):

- A DHCP server responds with a **DHCP Offer**, proposing an IP address and configuration settings to the client.

3. R (Request):

- The client sends a **DHCP Request** to the server, indicating acceptance of the offered IP address and configurations.

4. A (Acknowledge):

- The server sends a **DHCP Acknowledgment**, finalizing the lease of the IP address and other configurations to the client.

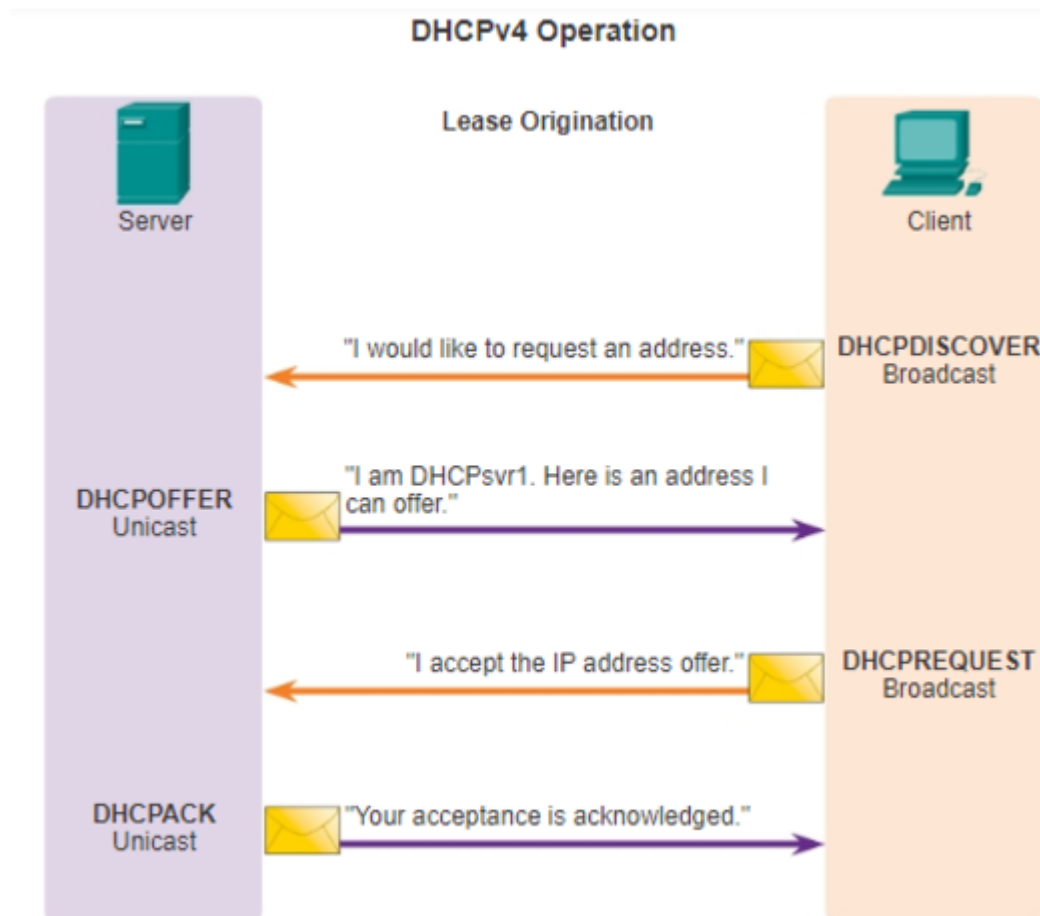
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APIPA

Automatic Private IP addressing with this, A DHCP client can automatically configure an IP address & subnet mask when no DHCP server is available.

It was a random address ranging from Class B from 169.254.0.1 to 169.254.255.254. default subnet mask of 255.255.0.0

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Unicast:

It is a one-to-one communication technique where communication occurs between a single client and a single recipient in a network.

For Example: Browsing any website

Multicast:

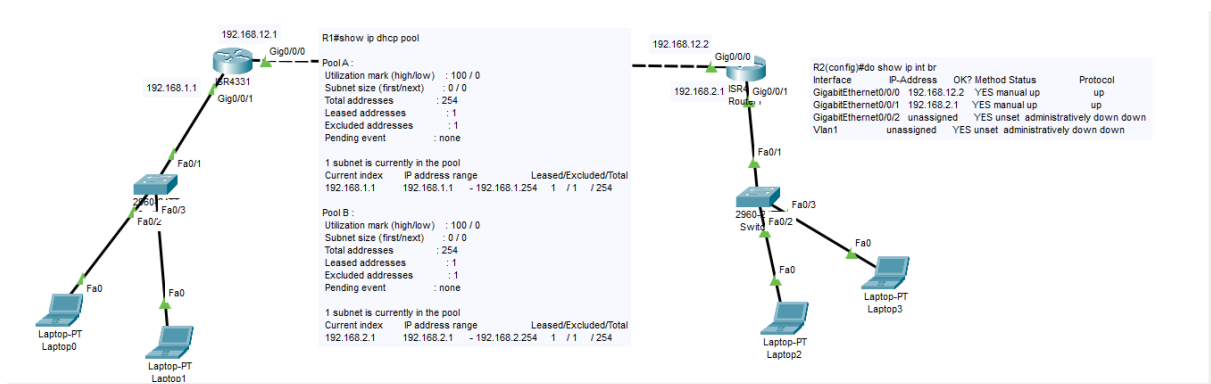
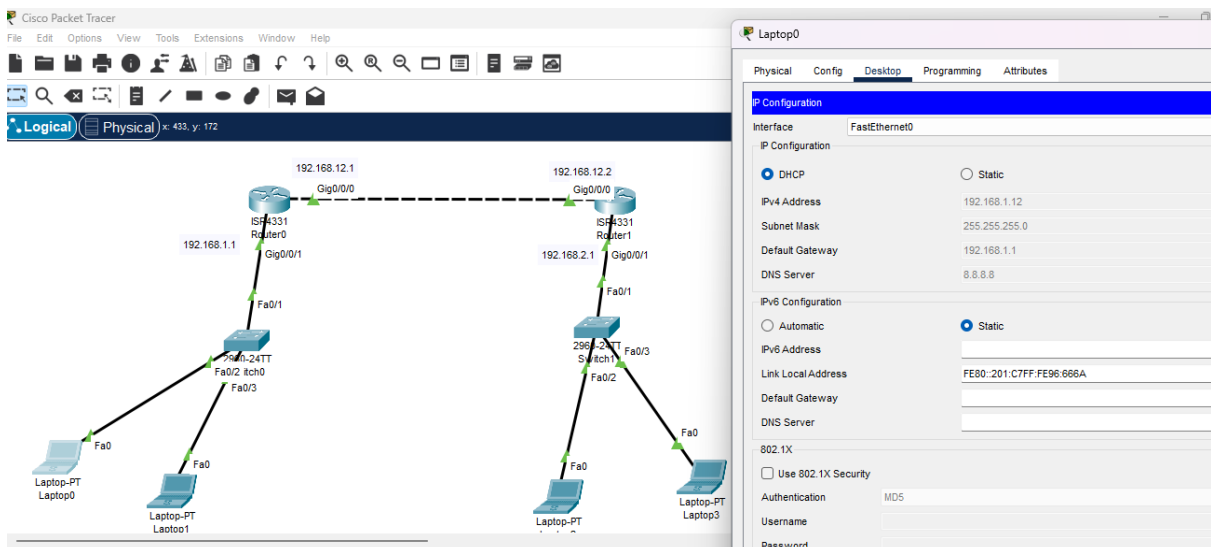
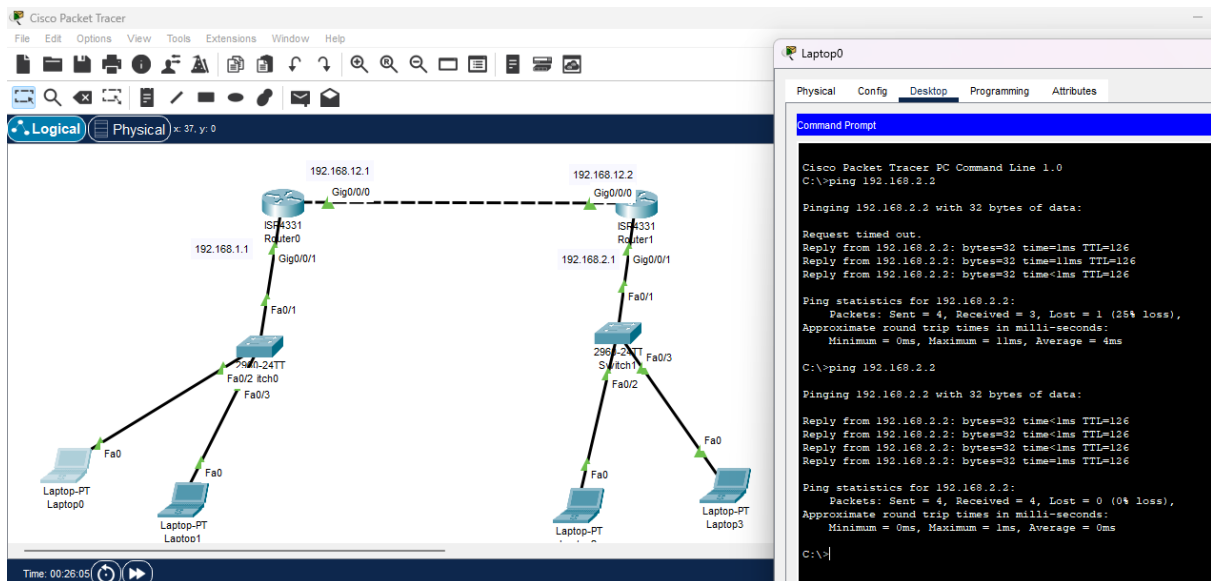
Multicast is a group type of communication where one or more senders transfer data to receivers. It uses the IGMP (Internet Group Management Protocol) to determine the receiver group.

For Example: Sending an Email to a particular group

Broadcast:

Broadcast is a one-to-many type of communication where data communication occurs between all the devices in the network. In broadcast, data is sent by one sender once and a copy of that data is available to all the devices in that network.

For Example: Television



R1

enable

```

conf t
hostname R1
int gig0/0/1
ip add 192.168.1.1 255.255.255.0
no sh

int gig0/0/0
ip add 192.168.12.1 255.255.255.0
no shutdown

ip route 192.168.2.0 255.255.255.0 192.168.12.2

ip dhcp pool A
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8

ip dhcp pool B
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
dns-server 8.8.8.8

ip dhcp excluded-address 192.168.1.1 192.168.1.10

##### R2 #####

enable
conf t
hostname R2
int gig0/0/0
ip add 192.168.12.2 255.255.255.0
no sh

int gig0/0/1
ip add 192.168.2.1 255.255.255.0
no shutdown

```

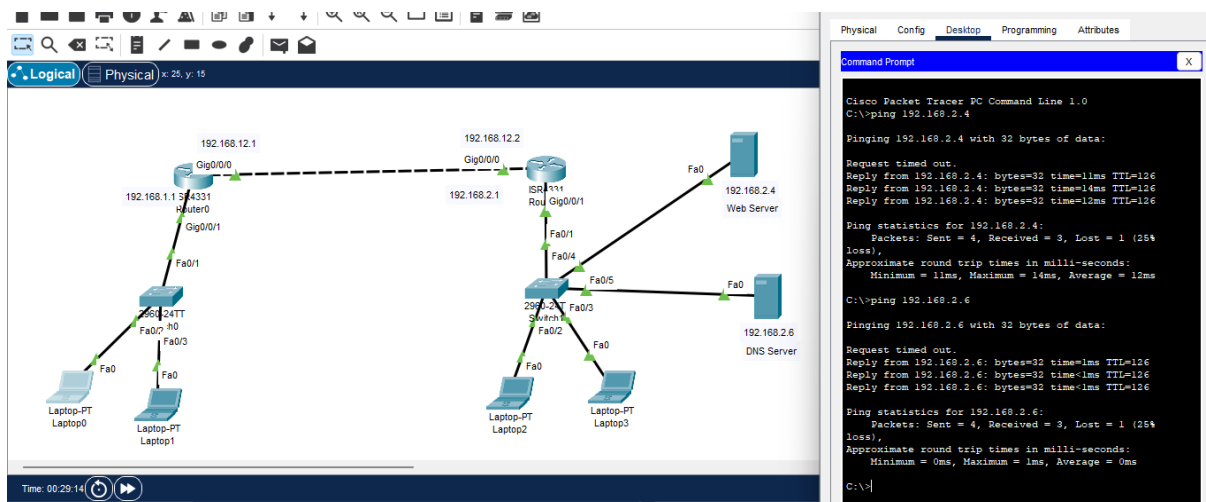
```
ip route 192.168.1.0 255.255.255.0 192.168.12.1

int gig0/0/1
ip helper-address 192.168.12.1
ip helper-address 192.168.13.1
```

3. DNS (Domain Name System):---

DNS is known as the phonebook that helps in translating the domain into a computer-readable IP address. DNS allows users to access websites without having to memorize long strings of numbers.

It acts as a directory for the internet, allowing devices to locate and connect to websites or services without needing to remember complex IP addresses.



Key Components of DNS

- DNS Servers:** These are specialized servers that store mappings of domain names to IP addresses.
 - Authoritative DNS Servers:** Provide the definitive IP address for a domain.
 - Recursive DNS Servers:** Handle client queries by contacting other DNS servers if needed.
- DNS Records:** Various record types are stored in DNS servers, such as:

- **A Record:** Maps a domain to an IPv4 address.
 - **AAAA Record:** Maps a domain to an IPv6 address.
 - **MX Record:** Identifies mail servers for a domain.
 - **CNAME Record:** Maps a domain to another domain (alias).
3. **DNS Resolver:** The client device or software (e.g., a router or computer) that initiates the DNS query.
4. **DNS Query Types:**
- **Recursive Query:** The DNS server handles the entire process of resolving the domain name.
 - **Iterative Query:** The client handles the query step-by-step by contacting multiple DNS servers.